

STORM User Guide

Severe Thunderstorm Observation and Reconnaissance Monitor

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1. Overview

STORM is a standalone desktop application purpose-built for severe weather storm chasing situational awareness. It integrates offline vector maps, real-time NEXRAD radar, GOES satellite imagery, SPC/NWS

hazard products, collaborative field annotations, meteorological drawings, and live multi-vehicle tracking — all in a single interface designed to function reliably in the field.

Core Technology Stack:

Component	Technology
UI Framework	PyQt6 (QMainWindow + QWebEngineView)
Map Renderer	MapLibre GL JS (embedded in browser engine)
Vector Tiles	Offline MBTiles served via custom <code>storm://</code> URL scheme
Radar Decoding	MetPy + matplotlib (Level 3, Unidata THREDDS)
Satellite Imagery	IEM WMS (GOES-East, GOES-West)
Hazard Data	SPC GeoJSON MapServer + NWS API
Vehicle/Annotation Sync	MQTT over AWS IoT (TLS)
Python-to-JS Bridge	QWebChannel (<code>bridge</code> object)

Run Command:

```
python main.py
```

2. Installation & Requirements

Prerequisites

- Python 3.10+
- PyQt6 with WebEngine (`pip install PyQt6 PyQt6-WebEngine`)
- MetPy, matplotlib, numpy, scipy (`pip install metpy matplotlib numpy scipy`)
- paho-mqtt (`pip install paho-mqtt`)
- The offline MBTiles vector tile database (`storm.mbtiles`, ~500 MB–1 GB, distributed separately)

MQTT Certificates

Certain network features (vehicle sync, annotation/drawing sync) require AWS IoT TLS certificates. Place them at:

- `/aws/storm.pem` — CA certificate
- `/aws/storm.pem.crt` — device certificate
- `/aws/storm-private.pem.key` — private key

If certificates are missing, STORM starts in a degraded state — map, radar, satellite, and SPC/NWS hazard features remain fully functional, but MQTT-dependent sync features are disabled.

Single-Instance Guard

Only one instance of STORM can run per machine (an internal port is used as a lock). If a second launch is attempted, a warning dialog appears and the second instance exits immediately.

3. Launching STORM

Launch Dialog

Every time STORM starts (unless `--monitor` mode is passed on the command line), a configuration dialog appears before the main window opens.

Launch Mode

Choose one of three mutually exclusive roles for this session:

Mode	Description
VEHICLE	Full participant — publishes local obs (file watcher / GPS) and syncs annotations, drawings, and cones over MQTT. Requires a vehicle passphrase.
MONITOR	View-only participant — all map overlays and MQTT inbound work, but no local obs are published. Requires a monitor passphrase.
VIEWER	Fully offline / no-network mode — no MQTT, no obs publishing. Map, radar, satellite, and SPC/NWS hazards still function.
ARCHIVE	Replay a past session — enter a UTC date/time and STORM fetches and plays back historical radar, satellite, hazards, soundings, and MQTT vehicle positions synchronized to a central time controller. Requires a passphrase. See Section 8 .

Archive Config (ARCHIVE mode only)

When ARCHIVE is selected, the vehicle-specific fields (ID, icon, data directory) are hidden and replaced with:

Archive Date/Time A date/time picker (calendar popup + time fields) for choosing the UTC start time of the replay session. STORM will fetch historical data beginning at this timestamp. The most recent HRRR run prior to the selected time is used for model data; radar, satellite, and hazard products are fetched from archives nearest to the start time.

The archive start time is not saved between sessions — each archive launch requires an explicit selection.

Vehicle Config (VEHICLE mode only)

Vehicle ID Your vehicle's identifier on the network (e.g., "Chase1", "Mesonet-TX"). Pre-filled from the previous session; edit freely. The ID is published with every observation, annotation, and drawing you send to other vehicles.

Vehicle Icon Choose the icon that represents your vehicle on the map. Options: car, drone, mesonet, lidar, radar, hailcam.

Data Directory (Track A) Optional folder path for a local FOFS instrument logger. If left blank, file-watcher input is disabled for the session. Use **Browse** to pick a folder. The app scans for `YYYYMMDD.txt` and polls it every 10 seconds.

Data on Launch (collapsible)

A collapsible section lets you pre-select which data layers to enable automatically on startup:

- **SPC / NWS / RADAR** — multi-select toggle buttons to auto-enable those overlays
- **Satellite** — exclusive selector: OFF | CONUS | AUTO-MESO
- **Surface obs** — multi-select: OK MESONET | WTM | KS MESONET

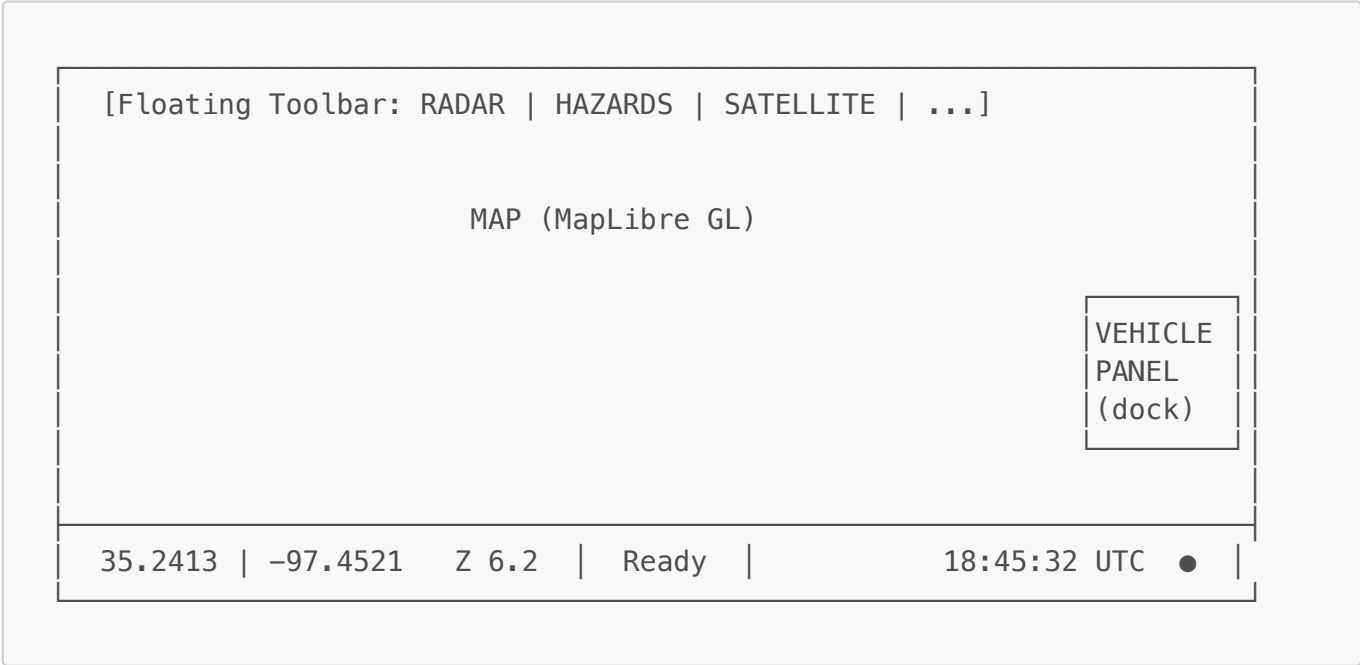
Update Status

During startup STORM checks for available updates. The dialog shows:

- "Checking for updates..." while polling
- Current and latest version numbers if an update is available
- An "Update Available" button (STORM never auto-updates)

Launch Click **LAUNCH** to proceed. Vehicle ID, icon, data directory, and mode are saved to persistent settings and pre-filled next time.

4. Main Interface Layout



Region	Description
Floating Toolbar	Top-left collapsible drawer strip — all overlay and tool controls
Map	Full-window interactive map; all overlays render here
Vehicle Panel	Right-side dock — vehicle list and selected-vehicle details

Region	Description
Outlook Panel	Left-side slide-out — SPC/NWS text discussion viewer
Status Bar	Bottom strip — mouse coordinates, zoom, status messages, clock, network indicator

5. Map Navigation

Mouse Controls

Action	Result
Scroll wheel	Zoom in / zoom out
Click + drag	Pan the map
Click a feature	Open info popup (SPC outlook, watch, warning, annotation, etc.)
Right-click	Finish current drawing or measurement
Hover over feature	Show tooltip

Map Layers (Render Order, bottom to top)

1. Offline vector base map (terrain, streets, water bodies, labels)
2. SPC categorical outlook polygons (MRGL / SLGHT / ENH / MDT / HIGH)
3. SPC probabilistic outlook polygons (tornado / wind / hail probability shading + hatching)
4. NWS watches (county-polygon outlines)
5. SPC Mesoscale Discussions (polygon outline + numeric label)
6. NWS warnings (storm-polygon fills, phenom-specific colors)
7. GOES satellite imagery (optional, opacity-controlled)
8. NEXRAD radar overlay (semi-transparent PNG raster)
9. Surface obs station model circles (mesonet obs)
10. Station plot icons (MetPy-style, at vehicle positions)
11. Vehicle markers (colored dots + info popups)
12. Annotations (road closure / construction / flooding / downed-lines / debris markers)
13. Drawings (fronts, polylines, polygons)
14. Storm motion cones
15. Routing line (active route geometry)
16. Measurement tool geometry
17. Deployment location markers

6. Floating Toolbar

The floating toolbar lives in the top-left corner of the map window. Each button either toggles a layer directly or expands a collapsible drawer with additional controls. Drawers animate open/closed smoothly.

6.1 Radar

Button: **RADAR** (checkable toggle)

Clicking the button expands the radar drawer. Checking the checkbox inside enables radar fetching and display.

Drawer Controls

Site Selector (dropdown) Choose from 31 pre-configured NEXRAD sites (e.g., KTLX, KOUN, KVNXX, KFWS, etc.) or type a custom 4-letter site code. Changing the site immediately clears cached frames and begins fetching for the new site.

Product Selector (dropdown)

Option	NEXRAD Product	Description
REFLECTIVITY (SR)	NOB (fallback: NOQ → NOR)	Base reflectivity (dBZ)
VELOCITY	NOU	Base velocity (kt)

Show Data Checkbox Enables or disables radar fetching. When unchecked, radar is hidden from the map but site/product selections are preserved.

Playback Controls

Button	Action
⏮	Jump to oldest cached frame
⏪	Step back one frame
▶ / ⏸	Start / pause auto-playback loop (500 ms per frame)
⏩	Step forward one frame
⏭	Jump to most recent (live) frame
Timeline slider	Scrub to any cached frame by dragging
Time label	Shows current frame's UTC time (HH:MMZ)

How Radar Data Works

- **Source:** Unidata THREDDS Level 3 catalog (public, no authentication required)
- **Polling:** Every 2 minutes
- **Frame cache:** Up to 12 frames per product
- **First fetch:** Backfills 6 reflectivity frames and 6 velocity frames
- **Render pipeline:** Level 3 polar data decoded by MetPy → reprojected to lat/lon grid → rendered as RGBA PNG via matplotlib → base64-encoded → pushed to MapLibre as a raster layer
- **Grid size:** 512×512 by default (configurable via `--render-grid-size`)
- **Render time:** ~80–150 ms on modern hardware at default grid size

Disabling Radar

Pass `--disable-radar` at launch to hide the RADAR button entirely.

6.2 Hazards

Button: `HAZARDS` (checkable toggle)

Clicking the button expands the hazards drawer. The drawer contains two groups of controls.

SPC Products (mutually exclusive — only one active at a time)

Button	Layer
OUTLOOK	Day 1 categorical outlook (MRGL, SLGHT, ENH, MDT, HIGH)
TOR	Tornado probability (2% / 5% / 10% / 15% / 30% / 45% / 60%)
WIND	Wind probability (5% / 15% / 30% / 45% / 60%)
HAIL	Hail probability (5% / 15% / 30% / 45% / 60%)

Selecting one clears the others. Clicking the already-active button deactivates it (all SPC probability layers hidden).

Additive Overlays (independent, stack on top of each other)

Button	Layer
WATCHES	SPC Tornado and Severe Thunderstorm Watch county polygons
MDs	SPC Mesoscale Discussion polygons with number labels
NWS WARNINGS	Active NWS warnings (all VTEC phenomenons)

These can be enabled in any combination, independent of the SPC product selection.

Legend

When any hazard product is active, a compact color legend row appears below the drawer controls showing color swatches and probability labels. It animates in and out with the drawer.

Clicking Hazard Features

SPC Categorical / Tornado / Wind / Hail polygons:

- Click a polygon → popup with risk level and probability label
- Click "Read Discussion" in the popup → fetches full text from IEM/SPC and displays it in the [Outlook Panel](#)

Watches:

- Click a county-polygon → popup with watch number, type (Tornado or Severe Thunderstorm), and issuance details

- Click "Read" → fetches full SEL bulletin text and displays it in the Outlook Panel

Mesoscale Discussions:

- Click an MD polygon → popup with MD number
- Click "Read" → fetches full MD text from SPC and displays it in the Outlook Panel

NWS Warnings:

- Click a warning polygon → popup with product type, event number, and NWS headline
- Full description and instructions fetched from the NWS API and shown in the popup

Significant Area Hatching

SPC tornado, wind, and hail probability layers include significant (CIG/SIGN) areas as separate overlapping features rendered with gray fill plus hatch patterns:

Pattern	Meaning
Dashed diagonal lines	Conditional Intensity Group 1 (CIG1)
Solid diagonal lines	CIG2 or SIGN (significant tornado area)
Checkerboard	CIG3

Color Reference

SPC Categorical:

Category	Color
MRGL (Marginal)	Green
SLGHT (Slight)	Yellow
ENH (Enhanced)	Orange
MDT (Moderate)	Red
HIGH	Purple

Watches:

Watch Type	Color
Tornado Watch	Red (#FF0000)
Severe Thunderstorm Watch	Royal Blue (#4169E1)

NWS Warnings by VTEC Phenom:

Phenom	Event	Color
TO	Tornado Warning	Red (#FF0000)

Phenom	Event	Color
SV	Severe Thunderstorm Warning	Gold (#FFD700)
FF	Flash Flood Warning	Green (#00FF00)
FL	Flood Warning	Cyan-green (#00FF7F)
WS	Winter Storm Warning	Pink (#FF69B4)
BZ	Blizzard Warning	Orange-red (#FF4500)

Mesoscale Discussions: Magenta (#FF66CC) outline

Data Polling

Product	Interval	Cache TTL
SPC Categorical + Probabilities	15 min	15 min
SPC Watches	2 min	None (always fresh)
SPC MDs	2 min	None
NWS Warnings	2 min	None

When you toggle a product on within the cache window, data displays instantly from cache. After TTL expiry, a fresh fetch is triggered automatically.

6.3 Satellite

Button: **SATELLITE** (checkable toggle)

Clicking the button expands the satellite drawer.

Coverage Mode Selection

Button	Description
CONUS	Full continental US view (1600×800 px, updates every 5 min)
MESO-1	Mesoscale sector 1 (1024×1024 px, updates every 1 min)
MESO-2	Mesoscale sector 2 (1024×1024 px, updates every 1 min)

MESO-1 and MESO-2 buttons are disabled until the satellite system confirms those sectors are available. Hover over MESO-1 or MESO-2 buttons to preview the sector boundary on the map as a semi-transparent overlay.

Opacity Slider

Controls satellite image transparency from 0% (invisible) to 100% (fully opaque). Default is 70%. Useful for seeing underlying map features through the imagery.

Playback Controls

Identical layout to the radar playback row:

Button	Action
⏮	Jump to oldest frame
⏪	Step back one frame
▶ / ⏸	Start / pause auto-playback (500 ms per frame)
⏩	Step forward one frame
⏭	Jump to newest frame
Timeline slider	Scrub to any cached frame
Time label	Current frame UTC time (--:--Z if no data yet)

How Satellite Data Works

- **Source:** IEM GOES-East WMS (mesonet.agron.iastate.edu) serving current operational GOES satellite (GOES-19 as of 2025)
- **Request format:** WMS GetMap PNG
- **Frame cache:** Up to 10 frames per mode
- **CONUS polling:** Every 5 minutes; 10 frames backfilled on mode selection
- **MESO polling:** Every 1 minute once sectors are confirmed available
- **Memory usage:** CONUS ~10–20 MB; each MESO sector ~20–30 MB

6.4 Soundings

Button: SOUNDINGS (checkable toggle, expands drawer)

Select which data source to use for Skew-T log-P soundings. Only one source is active at a time. Deactivating the button resets to HRRR mode.

Button	Source	How to Trigger
HRRR	NCEP HRRR model via open-meteo (no API key)	Click anywhere on the map
OBS	Observed radiosondes via IEM RAOB API	Radiosonde station markers appear; click one
NSSL	NSSL CLAMPS DL truck soundings via THREDDS	CLAMPS markers appear when data is available; click one

In HRRR mode, a click on the map immediately starts fetching. In OBS and NSSL modes, station/truck markers are rendered on the map first; clicking a marker fetches and opens the sounding dialog.

See [Section 10](#) for full dialog documentation.

6.5 Surface Obs

Button: SURFACE (checkable toggle, expands drawer)

Displays live surface observation station models from up to three mesonet networks simultaneously.

Network Toggles (multi-select)

Button	Network	Coverage
OK MESONET	Oklahoma Mesonet (OU)	~120 stations across Oklahoma
WTM	West Texas Mesonet (TTU)	~50 stations across West Texas
KS MESONET	Kansas Mesonet (K-State)	~50 stations across Kansas

Any combination can be active at once. Each network polls independently every 5 minutes.

Station Model Display

Each station renders a compact model circle on the map showing:

Position	Data
Upper-left	Temperature (°F)
Lower-left	Dewpoint (°F)
Center	Wind barb
Right	Wind speed / direction

Toggle station models on or off with the **Show plots** checkbox.

6.6 Routing

Button: ROUTING (checkable toggle, expands drawer)

Provides turn-by-turn driving directions rendered on the map.

Setting Origin and Destination

- **Origin:** Type an address, lat/lon pair, or click **Use My Location** to use the current GPS position.
- **Destination:** Type an address or lat/lon pair, or click a point on the map while the destination field is focused.

Click **Get Directions** to fetch the route. The route geometry is drawn on the map as a colored line, and the directions list appears below with each maneuver's distance and turn arrow (↑, ←, →, ↖, ↗, ↘, etc.).

Auto-Re-Routing

If the vehicle drifts more than 100 m off the calculated route for 3 or more consecutive GPS fixes, STORM automatically re-fetches directions from the current position.

Arrival

When the vehicle is within 30 m of the destination, STORM shows an arrival notification and clears the route from the map.

Data Sources

Function	Service
Turn-by-turn routing	OSRM public demo server (router.project-osrm.org)
Address geocoding	Nominatim (nominatim.openstreetmap.org)

6.7 Annotations

Button: **ANNOTATIONS** (checkable toggle, expands drawer)

Annotations are field-condition markers placed on the map and synced to all vehicles over MQTT.

Annotation Types

Icon	Type	Color	Use For
×	Road Closure	Red (#E53935)	Blocked road — no through traffic
▲	Construction	Yellow (#FFD166)	Active road work
~	Flooded Road	Blue (#4A9EFF)	Standing water over road
⚡	Downed Power Lines	Yellow (#FFD166)	Electrical hazard on road
!	Road Debris	Orange (#FF6B35)	Debris field on road

Placing an Annotation

1. Click the desired annotation type button in the drawer — it highlights in that type's color.
2. Click any location on the map.
3. A dialog opens showing the clicked coordinates and an optional "Note" text field.
4. Add any relevant note (e.g., "Road completely blocked at intersection with CR-1234").
5. Click **Add to Map**.
6. The annotation marker appears on the map immediately and is published to all connected vehicles via MQTT.

Editing or Deleting an Annotation

1. Click an existing annotation marker on the map.
2. The same dialog opens, pre-filled with the existing note.
3. Edit the note or click **Delete** to remove it.

- 4. Click **Save** to confirm.

Changes and deletions are published to all connected vehicles.

6.8 Drawings

Button: DRAWINGS (collapsible drawer)

Drawings are meteorological features sketched on the map — synced to all vehicles over MQTT.

Front Types

Button	Front	Symbol	Color
Cold Front	Cold Front	Triangles pointing in direction of motion	Blue (#4A9EFF)
Warm Front	Warm Front	Semicircles pointing in direction of motion	Red (#E53935)
Stationary Front	Stationary Front	Alternating triangles + semicircles on one line	Blue + Red
Occluded Front	Occluded Front	Alternating triangles + semicircles, same side	Purple (#9C27B0)
Dryline	Dryline	Scalloped arcs above the line	Brown (#D4872E)

Custom Shapes

Button	Shape	Color
—	Polyline	White (#E8EAF0)
□	Polygon	White (#E8EAF0)

Drawing a Front or Shape

1. Click the front type or shape button — it highlights and the map enters drawing mode (crosshair cursor).
2. Click the map to place the first point, then continue clicking to trace the line.
3. **Finish:** Right-click anywhere or press **Escape**.
4. A dialog prompts for a title (optional for fronts; required for polylines and polygons; defaults to the type name if left blank).
5. The feature appears on the map with the appropriate symbol and a title label at its center.

Editing or Deleting a Drawing

1. Click an existing drawing on the map.
2. The dialog opens with the current title.
3. For fronts: an option to flip direction (reverses the symbol orientation) is available.
4. Edit the title, flip direction, or click **Delete**.
5. Click **Save** to confirm.

All changes are synced via MQTT.

Disabled If

`--disable-annotations` is passed at launch — this flag disables annotations, drawings, station plots, and the storm motion cone together.

6.9 Storm Motion Cone

Button: **CONE**

Places a storm motion cone on the map — a forward-propagating sector showing the expected storm track with a 20° angular spread and time rings.

Creating a Cone

1. Click the **CONE** button.
2. Click any location on the map (typically the current storm location).
3. A dialog opens with two fields:
 - **Speed (knots):** Expected storm forward speed. Default 35 kts. Range: 0–200.
 - **Heading (degrees):** Direction the storm is moving *toward*. Default 240° (toward the SSW). Range: 0–359.
4. Click **Add to Map**.

The rendered cone includes:

- A base point at the clicked location
- Two radial bounding lines extending 1 hour of travel at $\pm 20^\circ$ spread
- Concentric time rings at 0.25, 0.5, 0.75, and 1.0 hour
- A center spine line showing the primary motion direction

Color: Cyan (#00CFFF)

Editing or Deleting a Cone

1. Click the cone on the map.
2. The dialog shows the current speed and heading.
3. Update the values or click **Delete**.
4. Click **Save**.

Cones are synced via MQTT (`storm/cones/{id}` topic).

Disabled If

`--disable-annotations` is passed at launch.

6.10 Measurement Tool

Button: **MEASURE** (checkable toggle)

Measures great-circle distances on the map.

Usage

- 1. Click the **MEASURE** button — the map cursor changes to a crosshair.
- 2. Click a start point on the map — a dot appears.
- 3. Move the mouse — a preview line and live distance label follow the cursor.
- 4. Click an end point — the segment is fixed.
- 5. Continue clicking to measure additional segments (multi-segment path).
- 6. **Finish:** Right-click or press **Escape**.
- 7. Distance is shown in miles (e.g., "47.3 mi") with total path length.

Clear: A **Clear** button removes all measurement geometry from the map.

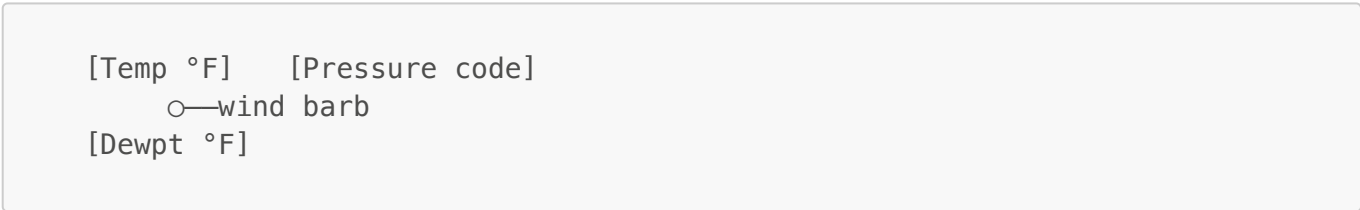
Color: Cyan lines and labels

6.11 Station Plots

Button: STATION PLOTS (checkable toggle)

Displays MetPy-style meteorological station plot icons at each tracked vehicle's current position.

Station Plot Layout



Position	Data	Color
Upper-left (NW)	Temperature (°F)	Red text
Lower-left (SW)	Dewpoint (°F)	Green text
Upper-right (NE)	Pressure code (last 3 digits × 10, e.g., 1013.2 mb → "132")	White text
Center	Wind barb (meteorological convention, barbs point toward station)	White
Center dot	Anchor	White circle

Wind Calm: If wind speed is ≤ 2 kt, only a small circle is rendered (no barbs).

How Station Plots Are Rendered

Station plots are generated as PNG images (135×135 px, transparent background) by MetPy and matplotlib, then pushed to the map as image markers. Each vehicle's plot is cached and only re-rendered when new observation data arrives.

Disabled If

`--disable-annotations` is passed at launch.

6.12 Deployment Locations

Button: `DEPLOY LOCS` (checkable toggle, expands drawer)

Displays historical deployment locations as markers on the map, filtered by research quality metrics.

Quality Metric Selector

Button	Metric	Description
<code>RANK_ABI</code>	ABI Rank	Rank by ABI-derived quality index
<code>RANK_AOI</code>	AOI Rank	Rank by area-of-interest quality
<code>RQI</code>	Road Quality Index	Continuous road surface quality score (0–1)

Threshold Slider

A slider controls the minimum quality threshold. Only locations meeting or exceeding the threshold are shown. The legend below the slider shows a green → red color ramp corresponding to the quality range, updated live as you drag.

Marker Radius

An additional slider controls the displayed circle radius for each location marker.

Data Source

- **File:** `data/deployment_locations.json`
- Markers are pre-loaded at startup; toggling visibility or threshold is instant.

Disabled If

`--disable-deploy-locs` is passed at launch, or the locations file is missing.

6.13 Vehicle Panel

Button: `VEHICLES` (toggles the right-side dock panel)

The vehicle panel provides a live roster of all tracked vehicles and time-series observation graphs.

Vehicle List (upper section)

Each vehicle row shows:

Field	Description
<code>ID</code>	Vehicle identifier (e.g., "Chase1")

Field	Description
Position	Compact lat/lon (e.g., "35.24
Last Update	Relative time (e.g., "just now", "2 min ago")
Color dot	Matches the vehicle's marker color on the map

Click a vehicle row to **zoom the map to that vehicle's location**.

Vehicles are sorted by most-recently-updated first.

7. Status Bar

The status bar runs along the bottom of the main window.

Section	Content
Left	Mouse coordinates (lat/lon) and zoom level, updated live as you move the cursor over the map. Example: 35.2413 -97.4521 Z 6.2
Center	Status messages: "Ready" (default), "MQTT connected", error messages (e.g., "Radar fetch error: timeout")
Right	Current UTC clock (updates every second). Example: 18:45:32 UTC
Far right	Network status indicator pill: Green = Online, Yellow = Slow (>1 s response), Red = Offline. Checked every 30 seconds via Cloudflare DNS.

8. Archive Mode

Archive mode lets you replay any past session at a chosen UTC date and time. All data layers — radar, satellite, hazards, soundings, and vehicle positions — are fetched from historical archives and driven by a central time controller.

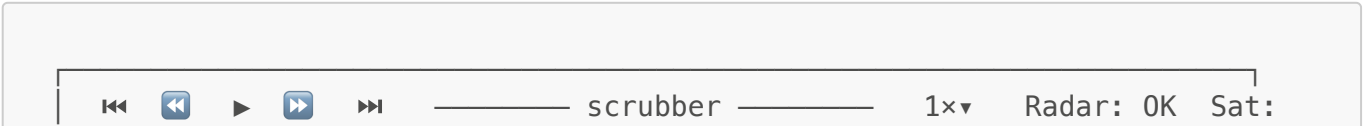
Entering Archive Mode

Select **ARCHIVE** in the launch dialog, enter the passphrase, and choose a UTC start date/time using the calendar picker. Click **LAUNCH**. A loading dialog appears while STORM prefetches the initial data for each layer before playback begins.

The window title and status pill display the current archive timestamp while in this mode: [ARCHIVE YYYY-MM-DD HH:MMZ].

Archive Controls Bar

A controls bar appears at the top of the map window in archive mode (it is not visible in live modes).



OK |

Playback Buttons

Button	Action
⏮	Jump to the session start time
⏪	Step back 30 seconds of archive time
▶ / ⏸	Start / pause automatic playback
⏩	Step forward 30 seconds of archive time
⏭	Jump to the current real-world time (end of archive)

Timeline Scrubber

Drag the slider to jump to any time within the archive session. The scrubber position represents seconds elapsed since midnight UTC of the session date.

Speed Selector

A dropdown next to the scrubber controls the playback speed multiplier:

Setting	Meaning
1×	Real-time (1 second of wall clock = 1 second of archive time)
5×	5 seconds of archive time per wall-clock second
10×	—
30×	—
60×	—
120×	—
300×	5 minutes of archive time per wall-clock second

Layer Status Indicators

Indicator	What It Shows
Radar:	Current radar fetch status ("OK", "waiting", "no data")
Sat:	Current satellite fetch status

These update as each fetcher resolves data for the current archive timestamp.



How Archive Data Works

Layer	Source	Behavior
NEXRAD Radar	Unidata THREDDS archive	Fetches the Level 3 scan nearest to the current archive time
GOES Satellite	IEM WMS historical endpoint	Fetches the frame nearest to the current archive time
SPC/NWS Hazards	Archived GeoJSON products	Fetches once at session start for the session date
Soundings	open-meteo HRRR archive	Fetches on demand (map click), using archive time as valid time
Vehicle Positions	MQTT message log	Historical vehicle obs replayed in time order

Each layer updates automatically as the archive clock advances. Data fetches are triggered when the clock crosses boundaries (e.g., every radar scan interval).

Limitations in Archive Mode

- Vehicle positions are read from stored MQTT message logs — only vehicles that were active and publishing during the original session will appear.
- Real-time MQTT sync (annotations, drawings) is inactive in archive mode.
- Local file watcher (Track A) and GPS reader (Track B) are inactive.
- Network connectivity is still required — archive data is fetched from remote servers at playback time.

9. Outlook Text Panel

The Outlook Panel is a right-side panel that expands horizontally to display the full text of SPC and NWS products.

Opening the Panel: Click any SPC feature (outlook area, watch polygon, or MD polygon) on the map, then click "Read" or "Read Discussion" in the popup.

What Appears:

- **Title bar:** Product name (e.g., "DAY 1 CONVECTIVE OUTLOOK", "MD 0179", "TORNADO WATCH 0029")
- **Close button (✕):** Collapses the panel
- **Text area:** Full monospaced text of the product (read-only, scrollable)

Text Sources:

Product	Source
Day 1 Outlook	IEM AFOS (pil=SWODY1)

Product	Source
Mesoscale Discussions	SPC direct (spc.noaa.gov/products/md/md{NNNN}.txt)
Tornado Watches	IEM AFOS (pil=SEL{n} where n = watch_number % 10)
SVR Tstm Watches	IEM AFOS (same PIL logic as tornado)
NWS Warnings	NWS API JSON (headline + description + instruction fields)

Animation: Expands in from the right edge (200 ms), collapses on close (200 ms). Panel is 340 px wide.

10. Point Soundings

STORM supports three independent sounding sources, selected via the **SOUNDINGS** drawer (see [Section 6.4](#)). All three sources display in the same Skew-T log-P dialog.

HRRR Model Soundings

Click any map location within the HRRR CONUS domain. Data fetched from open-meteo (free tier, no API key required). The dialog opens automatically when data arrives (typically 1–3 seconds). Clicking a new location updates the same dialog in place.

Observed Radiosonde Soundings

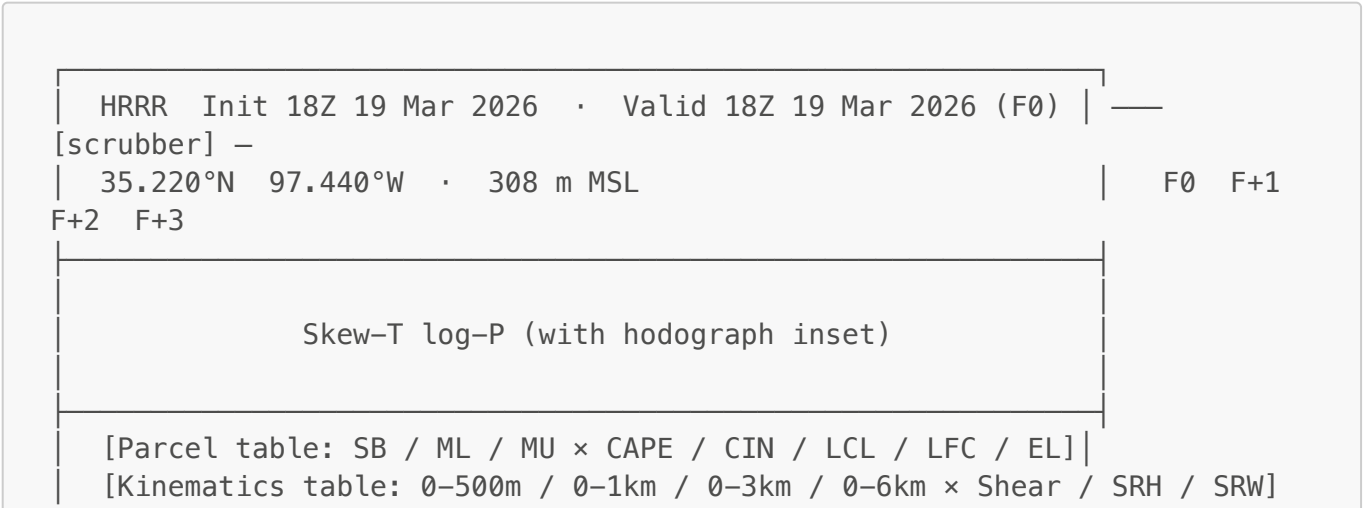
Switch to OBS mode in the SOUNDINGS drawer. Radiosonde station markers appear on the map at active upper-air sites. Click a marker to fetch the most recent balloon launch profile from IEM RAOB. STORM automatically selects the 00Z or 12Z launch closest to the current UTC time.

NSSL CLAMPS DL Truck Soundings

Switch to NSSL mode in the SOUNDINGS drawer. CLAMPS truck markers appear when data is available from the NSSL THREDDS fileserver. Click a marker to fetch and display the most recent truck sounding. Up to 12 hours of soundings per truck are available.

Triggering a Sounding

Dialog Layout





Header & Scrubber

The header shows the HRRR init time and the currently-displayed valid time. The scrubber on the right side steps through four time slots: **F0** (analysis), **F+1h**, **F+2h**, and **F+3h** from the most recent HRRR run. Drag the slider or click a tick label to switch hours. The active slot is highlighted in cyan.

Skew-T diagram

Element	Description
Red curve	Temperature
Green curve	Dewpoint
Dashed warm-tint curve	Virtual temperature
White dashed curve	Surface-based parcel profile
Red shading	CAPE area
Blue shading	CIN area
Cyan circle	LCL marker
Blue shading (light)	Dendritic growth zone (−10 to −20°C)
Red dotted horizontal lines	AGL height reference lines (0.5, 1, 2, 3, 4, 6, 9 km)
Green bracket (left spine)	Effective inflow layer (EIL)
Wind barbs	Meteorological wind at each pressure level

Hodograph Inset

Located in the upper-right corner of the Skew-T. The trace is color-coded by height AGL:

Color	Layer
Red	0–3 km
Gold	3–6 km
Blue	6–9 km
Gray	Above 9 km

A wide semi-transparent green overlay highlights the EIL segment. Bunkers right-mover (RM) and left-mover (LM) storm motion points are plotted as colored dots (red and blue respectively) with direction/speed readouts in the corner of the inset.

Parcel Table

Column	Description
CAPE	Convective available potential energy (J/kg)
CIN	Convective inhibition (J/kg, negative values)
LCL	Lifted condensation level height (m AGL)
LFC	Level of free convection (m AGL)
EL	Equilibrium level (m AGL)

Rows: **SB** (surface-based), **ML** (100 hPa mixed-layer), **MU** (most-unstable). Values are threshold-colored: yellow → orange → red as severity increases.

Kinematics Table

Row	Description
0–500m	Bulk shear, SRH, and mean SRW in the lowest 500 m
0–1km	Bulk shear, SRH, and mean SRW in the lowest 1 km
0–3km	Bulk shear, SRH, and mean SRW in the lowest 3 km
0–6km	Bulk shear and mean SRW (SRH not shown for this layer)

SRW (storm-relative wind) is the mean wind speed relative to the Bunkers right-mover in each layer. All shear values in knots, SRH in m²/s².

Storm motion is displayed as **RM dir°/spd kt** and **LM dir°/spd kt** in the hodograph corner.

Bottom Scalar Row

Parameter	Description
LR 700–500	700–500 hPa lapse rate (°C/km)
LR 0–3 km	0–3 km AGL lapse rate (°C/km)
SFC θe	Surface equivalent potential temperature (K)
PW	Precipitable water (mm)
Conv Temp	Convective temperature — surface temperature required for convection initiation (°C)
STP	Significant Tornado Parameter
SCP	Supercell Composite Parameter
EHI	Energy-Helicity Index

Cursor Readout

Hovering over the Skew-T axes displays a live readout below the plot: `pressure (hPa) · T · Td · Wind dir°@spd kt · height m MSL`

Data Sources

HRRR mode:

Item	Detail
Model	NCEP HRRR CONUS (3 km, hourly updates)
API	open-meteo /v1/forecast — no API key required
Variables	Temperature, dewpoint, U/V wind components, geopotential height at 25 pressure levels (1000–100 hPa)
Request cost	1 API call per map click
Rate limits (free tier)	600/min · 10,000/day · 300,000/month
Fetch time	Typically 1–3 seconds
Domain	CONUS only — clicks outside HRRR coverage will return an error

OBS mode:

Item	Detail
Source	IEM RAOB radiosonde archive
API	mesonet.agron.iastate.edu/json/raob.py
Launch times	00Z and 12Z; STORM selects the most recent launch
Fetch time	Typically 2–5 seconds

NSSL mode:

Item	Detail
Source	NSSL CLAMPS DL truck soundings via THREDDS fileServer
Catalog	data.nssl.noaa.gov/thredds/catalog/FRDD/CLAMPS/dltruck/
Availability	Only when NSSL DL truck is actively collecting data
History	Up to 12 hours of soundings per truck available

11. Vehicle Tracking & Observations

STORM tracks vehicles through several parallel data input channels. All inputs feed into the live vehicle state used by the vehicle panel, station plots, and map markers.

Track A — Local File Watcher

Polls a FOFS truck logger format file every 10 seconds. Enabled only if a data directory was selected at launch.

Expected filename: `YYYYMMDD.txt` (today's date)

Expected columns (tab or comma delimited):

Column	Format	Description
<code>gps_date</code>	DDMMYY	GPS date
<code>gps_time</code>	HHMMSS	GPS time (UTC)
<code>lat</code>	decimal degrees	Latitude
<code>lon</code>	decimal degrees	Longitude
<code>t_fast</code>	°F	Fast-response temperature
<code>dewpoint</code>	°F	Dewpoint temperature
<code>sfc_wspd</code>	knots	Surface wind speed
<code>sfc_wdir</code>	degrees	Surface wind direction
<code>pressure</code>	mb	Atmospheric pressure

Track B — GPS Reader

Automatically detects and reads NMEA sentences from a connected serial GPS device. Provides position-only data (no met obs). Active only if `--disable-data-inputs` is not passed.

MQTT — Inbound Vehicle Observations

Receives observations from other vehicles via the `storm/vehicles/{id}` topic. Active whenever MQTT is connected and not disabled.

12. Network & MQTT Sync

STORM uses MQTT over AWS IoT (TLS) to synchronize annotations, drawings, cones, and vehicle observations across all connected vehicles in the network.

Topics

Topic	Content
<code>storm/vehicles/{id}</code>	Vehicle observations (lat, lon, temp, dewpoint, wind, pressure)
<code>storm/annotations/{id}</code>	Road condition annotations (create / edit / delete)
<code>storm/drawings/{id}</code>	Meteorological drawings (fronts, polylines, polygons)
<code>storm/cones/{id}</code>	Storm motion cones

Behavior

- On connect, retained messages for all active annotations, drawings, and cones are received and displayed automatically.
- Disconnect/reconnect is handled automatically — STORM continuously attempts to reconnect with exponential backoff.
- If `--disable-mqtt` is passed, all sync features are disabled and the vehicle panel shows only locally-observed vehicles.
- Status bar shows "MQTT connected" or an error message.

13. Data Sources & Polling Intervals

Data Source	Polling Interval	Notes
NEXRAD Radar (Unidata THREDDS)	Every 2 minutes	Level 3 products; no auth required
SPC Categorical Outlook	Every 15 minutes	Day 1 only
SPC Tornado / Wind / Hail Probabilities	Every 15 minutes	All fetched together with categorical
SPC Watches	Every 2 minutes	County polygons, tornado + SVR tstm
SPC Mesoscale Discussions	Every 2 minutes	MD polygon geometry + number
NWS Warnings	Every 2 minutes	All active VTEC phenomenons
GOES Satellite (CONUS)	Every 5 minutes	Full CONUS, up to 10 frames
GOES Satellite (MESO-1/2)	Every 1 minute	When active, per-sector
HRRR Point Sounding (open-meteo)	On demand	1 API call per map click
Observed Radiosonde (IEM RAOB)	On demand	Nearest 00Z / 12Z launch
NSSL CLAMPS DL Truck	On demand	Fetched from NSSL THREDDS on click
OK / KS / WTM Mesonet (surface obs)	Every 5 minutes	Per-network, independent toggles
Turn-by-turn Routing (OSRM)	On demand	Re-fetched automatically if off-route
Local Obs File (Track A)	Every 10 seconds	Today's YYYYMMDD.txt
GPS (Track B)	Real-time / continuous	NMEA sentences
MQTT Inbound	Real-time	Vehicle obs, annotations, drawings, cones
Network Health Check	Every 30 seconds	Cloudflare 1.1.1.1 ping

14. Command-Line Options

Run `python main.py --help` for the full list. Key options:

General

<code>--debug</code> (Ctrl+D)	Enable debug logging and the debug panel
<code>--log-level LEVEL</code> ERROR (default: WARNING)	Logging verbosity: DEBUG, INFO, WARNING,
<code>--monitor</code>	Monitor mode: suppresses local obs inputs

Disable Features

<code>--disable-radar</code> fetching	Hide the RADAR control; disable all radar
<code>--disable-mqtt</code> obs)	Disable all MQTT features (sync, vehicle
<code>--disable-annotations</code> plots, and cones	Disable annotations, drawings, station
<code>--disable-deploy-locs</code>	Disable deployment location markers
<code>--disable-data-inputs</code>	Disable local file watcher and GPS reader

Replay Mode

<code>--truck-replay-file PATH</code> live data	Replay a local obs CSV/TXT file instead of
<code>--truck-replay-interval-ms MS</code> (default: 1000 ms)	Interval between replayed samples
<code>--truck-replay-restamp</code> "now"	Shift timestamps so the last obs lands at

Performance

<code>--render-grid-size N</code> (default), 768, or 1024	Radar rendering grid: 128, 256, 512
--	-------------------------------------

Debug Run Profiles

<code>--debug-run N</code> size)	Quick diagnostic profile (0–6): 0 = full features 1 = disable MQTT only 2 = safe mode (minimal features, low grid
<code>--enable-startup-toggles</code> debug-run profile	3–6 = various feature subsets for testing Allow CLI disable flags to override the

Network

`--mqtt-no-tls`

MQTT without TLS (development use only, not for field use)

15. Configuration & Certificates

config.py

Edit `config.py` for persistent defaults:

Constant	Description
<code>ACCENT_COLOR</code>	UI accent color (default: <code>"#00CFFF"</code> , cyan)
<code>VEHICLE_ID</code>	Default vehicle ID (overridden by launch dialog)
<code>OBS_FILE_DIR</code>	Default data directory path (overridden by launch dialog)
<code>MQTT_HOST</code>	AWS IoT endpoint hostname
<code>MQTT_PORT</code>	MQTT port (default: 8883 for TLS)
<code>MQTT_USE_TLS</code>	Whether to require TLS (default: True)

Persistent Session State (QSettings)

The following are saved automatically and restored at next launch:

- Window geometry and dock layout
- Last-used vehicle ID
- Last-used data directory

Settings are stored under the `"NSSL"` / `"STORM"` namespace in your OS's native settings storage.

TLS Certificate Files

Place these three files exactly at these paths before using MQTT:

File	Role
<code>/aws/storm.pem</code>	CA certificate
<code>/aws/storm.pem.crt</code>	Device certificate
<code>/aws/storm-private.pem.key</code>	Private key

Missing certificates cause MQTT to fail silently — all other features remain operational.

16. Keyboard Shortcuts

Key	Action
Escape	Cancel active drawing / annotation placement / measurement mode
Ctrl+D	Toggle debug panel
Ctrl+E	Toggle error log panel
Right-click	Finish drawing or measurement in progress

17. Performance Tuning

Radar Grid Size

The radar rendering grid controls the sharpness and rendering time of the NEXRAD overlay.

Grid Size	Render Time (approx.)	Recommended Use
128	~5–10 ms	Very slow hardware, emergency fallback
256	~20–40 ms	Older laptops, safe-mode deployments
512	~80–150 ms	Default — balanced quality/speed
768	~200–400 ms	High-end hardware, sharper image
1024	~400–800 ms	Workstation-class machines only

Pass `--render-grid-size N` to override. Lower grid sizes make the radar appear blockier but keep the CPU free for other tasks.

Memory Usage

Component	Approximate Memory
Radar frame cache (12 frames)	20–50 MB
CONUS satellite cache (10 frames)	10–20 MB
MESO satellite cache (10 frames/sector)	20–30 MB per sector
Station plot image cache	~1–5 MB (per vehicle)
Observation history buffer (10 min/vehicle)	Negligible

Safe Mode

If the application is unstable on older hardware (WebGL context loss, slow renders), launch with:

```
python main.py --debug-run 2
```

Safe mode reduces grid size to 128, disables some overlays, and minimizes background polling.

18. Diagnostics & Error Handling

Debug Mode

```
python main.py --debug
```

Enables:

- Verbose logging to console (DEBUG level)
- The debug panel (also toggled with **Ctrl+D** at any time)
- Fault handler writing stack traces to `storm_fault.log` on crash

Error Log Panel

Press **Ctrl+E** at any time to open the error log panel. It shows all WARNING-level and above log messages captured since launch. This log is also written to `storm_errors.log` and survives across restarts.

Network Status Indicator

The colored pill in the bottom-right of the status bar updates every 30 seconds:

- **Green (Online):** Connectivity to Cloudflare 1.1.1.1 is fast
- **Yellow (Slow):** Response time exceeded 1 second — data fetches may be delayed
- **Red (Offline):** No connectivity — no real-time data will update

Single-Instance Guard

STORM uses an internal TCP port (19876) as a process lock. If a second instance is launched while one is already running, a dialog warns the user and the second instance exits.

Common Issues

Symptom	Likely Cause	Resolution
Radar not updating	THREDDS server unavailable or slow	Check network indicator; try a different site
"MQTT disconnected" in status bar	Certificate missing or endpoint unreachable	Verify <code>/aws/</code> certificate files exist
Station plots missing	<code>--disable-annotations</code> passed	Relaunch without that flag
MESO-1/2 buttons grayed out	Sectors not currently available	Wait — buttons enable automatically when sectors are active

Symptom	Likely Cause	Resolution
Map tiles blank/gray	<code>storm.mbtiles</code> missing or corrupted	Verify the MBTiles file is present and not zero-byte
Slow/choppy radar rendering	Grid size too large for hardware	Use <code>--render-grid-size 256</code>
Hatch patterns not visible	WebGL limitation	Known quirk; patterns are built as pixel data, not canvas — should render in all supported modes

19. Known Limitations & Quirks

Canvas Readback MapLibre GL running inside QWebEngineView cannot reliably read back canvas pixel data. Hatch patterns (CIG/SIGN significant areas) are therefore built entirely as raw `Uint8Array` pixel data and added to MapLibre via `addImage()`, rather than drawn on a canvas. This is a known architectural constraint.

QWebChannel Async Initialization The `bridge` object that connects Python and JavaScript is initialized asynchronously after the map loads. All JavaScript API functions are stubbed as no-ops (`_stormNoop`) until the bridge is ready. In practice this is transparent to the user, but very rapid actions in the first second after launch may be silently dropped.

MBTiles File Not Included in Repository The offline vector tile database (`storm.mbtiles`) is large (500 MB–1 GB) and is not included in the git repository. It must be downloaded separately and placed in the project root.

SPC Text Product Latency IEM AFOS sometimes responds slowly (up to 20 seconds per request). STORM waits patiently rather than timing out aggressively. This is normal behavior during high-traffic SPC issuance periods.

MESO Sectors Availability GOES MESO-1 and MESO-2 sectors are not always active or available over your area of interest. STORM automatically enables/disables the MESO buttons based on sector availability. This is a function of NOAA's operational GOES scanning schedule.

GOES Satellite Band The IEM WMS endpoint serves the default operational visible or IR band for the current GOES satellite. Band selection is not user-configurable from within STORM.

SPC Data Lag SPC GeoJSON products (tor, wind, hail) are typically updated once or twice daily. They may not reflect very recent issuances until the next polling cycle completes (up to 15 minutes after the fact). SPC Watches and MDs poll every 2 minutes and are effectively near-real-time.

20. Feature Availability Matrix

Feature	Default	Disabled By
Offline vector base map	✅ Always	n/a (requires storm.mbtiles)
NEXRAD Radar overlay	✅ Enabled	<code>--disable-radar</code>

Feature	Default	Disabled By
GOES Satellite overlay	✓ Enabled	n/a
SPC Hazards (outlook, tor, wind, hail)	✓ Enabled	n/a
SPC Watches	✓ Enabled	n/a
SPC Mesoscale Discussions	✓ Enabled	n/a
NWS Warnings	✓ Enabled	n/a
Outlook Text Panel	✓ Enabled	n/a
HRRR Point Soundings	✓ Enabled	n/a
Observed Radiosonde Soundings	✓ Enabled	n/a
NSSL CLAMPS Truck Soundings	✓ Enabled (when data available)	n/a
Surface Obs (OK / KS / WTM Mesonet)	✓ Enabled (when toggled on)	n/a
Turn-by-turn Routing	✓ Enabled	n/a
Annotations (road conditions)	✓ Enabled	<code>--disable-annotations</code>
Drawings (fronts, polylines, polygons)	✓ Enabled	<code>--disable-annotations</code>
Storm Motion Cone	✓ Enabled	<code>--disable-annotations</code>
Station Plots (vehicle obs)	✓ Enabled	<code>--disable-annotations</code>
Measurement Tool	✓ Enabled	n/a
Deployment Locations	✓ Enabled	<code>--disable-deploy-locs</code> or missing data file
Vehicle Panel	✓ Enabled	n/a
MQTT Sync (annotations, drawings, cones)	✓ Enabled	<code>--disable-mqtt</code> or <code>--disable-annotations</code>
MQTT Vehicle Obs (inbound)	✓ Enabled	<code>--disable-mqtt</code>
Local File Watcher (Track A)	✓ Enabled (if dir set)	<code>--disable-data-inputs</code> or no dir selected
GPS Reader (Track B)	✓ Enabled (if device present)	<code>--disable-data-inputs</code>
Radar Playback (loop mode)	✓ Enabled	<code>--disable-radar</code>
Satellite Playback (loop mode)	✓ Enabled	n/a

Feature	Default	Disabled By
Archive Mode (session replay)	 Enabled	Requires passphrase; selected at launch
Archive Radar Playback	 Enabled in archive mode	n/a
Archive Satellite Playback	 Enabled in archive mode	n/a
Archive Hazard Products	 Enabled in archive mode	n/a
Archive Vehicle Positions (MQTT log)	 Enabled in archive mode	n/a
Archive Soundings	 On demand in archive mode	n/a

STORM — Severe Thunderstorm Observation and Reconnaissance Monitor Internal field operations tool — NSSL Mobile Mesonet program